

Tide retrospective: projective agreement fixes its own gauge equal zero loci, scalar rigidity, and the $1/Z$ closure

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2026-09-19

Abstract

Two nonnegative losses whose Laplace families agree projectively beyond all orders, $\int \phi e^{-tL_2} - C(t) \int \phi e^{-tL_1} = o(t^{-\infty})$ for every $\phi \in C_c^\infty$ and an arbitrary scalar function C , have the same zero set as soon as both have a zero; the scalar is polynomially tame and eventually positive; and once the losses agree near one zero, which the morning's germ theorem supplies at any common analytic zero, $C(t) = 1 + o(t^{-\infty})$ and the two families agree *exactly* beyond all orders at every tested observable. With $C = Z_2/Z_1$ this closes the germbij note's normalized story: equality of the normalized expansions $\Phi_{L_1} = \Phi_{L_2}$ for smooth nonnegative losses analytic at their zeros gives equal zero sets, local equality on an open neighbourhood of the zero set, $Z_2/Z_1 = 1 + o(t^{-\infty})$, and exact agreement of every moment. The load-bearing new lemma is the anchor, $\int \psi e^{-tK} \geq \kappa t^{-d/2}$ near a C^2 zero, the $a \equiv 1$ case of the seabed's sector bound; it discharges a polynomial lower bound that three seabed theorems had carried as a hypothesis since the anchoring tide. Six hundred and fifty lines, two LSP rounds, no analyticity anywhere except through the germ theorem.

1 Setting

The morning's tide, [2026-09-19-06-05](#), proved that projective agreement forces $L_1 = L_2$ near every common analytic zero, whatever C is, and recorded two stretch corollaries from its consult: that equal zero loci are forced without analyticity, and that the scalar must be $1 + o(t^{-\infty})$. Both rest on one missing estimate. The seabed's scalar-gauge algebra in `Anchoring` and `OnePointAnchoring` (`superPoly_of_mul_anchor`, `superPoly_sub_of_scalar_gauge`, `anchored_proportionality_remove_scalar`, `one_point_anchoring_contradiction`) is complete modulo a hypothesis `hanchor_low`: a positive polynomial lower bound on the reference moment $\int \psi e^{-tL_1}$ at an anchoring bump. Nothing in the seabed proved such a bound, although the sector lower bound `sector_lower_bound_multi` contains it as the degenerate case $a \equiv 1$, $m = 0$. The deliberation (tide log `tide-log/2026-09-19-15-40-tide-germbij-projective-closure.md`, consult in `gpt_germbij_projective_v1.md`) settled on packaging that anchor, the complementary exponential gap bound away from the zero set, and the four consequences: tameness of C , equal zero loci, scalar rigidity with the transfer to exact agreement, and the $1/Z$ headline. Both parties voted for the whole package.

2 The thing formalised

Write $I_i(\phi, t) = \int \phi e^{-tL_i}$ and call the families *projectively agreeing* (**ProjectiveAgreement** L1 L2 C) when $I_2(\phi) - C I_1(\phi) = o(t^{-\infty})$ for every smooth compactly supported ϕ .

- (G) If $K \geq 0$ is continuous, $K(p) = 0$, K is C^2 at p , and $\psi \geq 0$ is continuous, compactly supported, and equal to 1 near p , then $\int \psi e^{-tK} \geq \kappa t^{-d/2}$ for large t .
- (E) If $L \geq \delta > 0$ on the support of ψ , then $|\int \psi e^{-tL}| \leq \|\psi\|_1 e^{-t\delta}$ for $t \geq 0$, and $B(t) \int \psi e^{-tL} = o(t^{-\infty})$ for any polynomially bounded B .
- (T) For continuous nonnegative L_1, L_2 agreeing projectively: a C^2 zero of L_1 gives $|C(t)| \leq A t^d$ eventually; a C^2 zero of L_2 gives $C(t) \geq c t^{-d} > 0$ eventually.
- (Z) If moreover both losses are C^2 near their zeros and both have a zero, their zero sets coincide.
- (R) If $L_1 = L_2$ near some C^2 zero p of L_1 , then $C(t) = 1 + o(t^{-\infty})$, and $I_2(\phi) - I_1(\phi) = o(t^{-\infty})$ for every tested ϕ .
- (N) For smooth nonnegative losses analytic at their zeros, both with zeros, and a nonnegative window χ with positive partition values: if the normalized families agree beyond all orders for every tested ϕ , then the zero sets coincide, $L_1 = L_2$ on an open neighbourhood of the zero set, $Z_2/Z_1 = 1 + o(t^{-\infty})$, and $I_2(\phi) - I_1(\phi) = o(t^{-\infty})$ for every tested ϕ .

The Lean signature of the headline:

```
theorem normalized_expectations_closure {L1 L2 chi : (iota -> R) -> R}
  (h1 : ContDiff R infy L1) (h2 : ContDiff R infy L2)
  (hL1 : forall w, 0 <= L1 w) (hL2 : forall w, 0 <= L2 w)
  (hA1 : forall p, L1 p = 0 -> AnalyticAt R L1 p)
  (hA2 : forall p, L2 p = 0 -> AnalyticAt R L2 p)
  (hne1 : exists p, L1 p = 0) (hne2 : exists q, L2 q = 0)
  (hchic : Continuous chi) (hchis : HasCompactSupport chi)
  (hchi0 : forall w, 0 <= chi w)
  (hZ1 : forall t, 0 < integral (chi * exp (-(t * L1))))
  (hZ2 : forall t, 0 < integral (chi * exp (-(t * L2))))
  (hfam : forall phi, ContDiff R infy phi -> HasCompactSupport phi ->
    SuperPoly fun t => I2 phi t / Z2 t - I1 phi t / Z1 t) :
  (forall w, L1 w = 0 <-> L2 w = 0) /\
  (exists U, IsOpen U /\ {w | L1 w = 0} <= U /\
    forall w in U, L1 w = L2 w) /\
  SuperPoly (fun t => Z2 t / Z1 t - 1) /\
  forall phi, ContDiff R infy phi -> HasCompactSupport phi ->
    SuperPoly fun t => I2 phi t - I1 phi t
```

(integrals abbreviated). The components are `exists_lower_bound_integral_exp_near_zero`

and `anchor_lower_bound_eventually` for (G), `abs_integral_mul_exp_le_of_gap` and `superPoly_polyBounded_mul_integral_of_gap` for (E), `exists_scalar_upper_bound` and `eventually_scalar_lower_bound` for (T), `zero_iff_zero_of_projective` for (Z), `superPoly_scalar_sub_one_of_eventuallyEq`, `superPoly_difference_of_projective` and `projective_forces_exact_at` for (R).

Nothing is assumed of C : not a sign, not measurability, not growth. Every estimate involving $C(t)$ is pointwise in t , and (T) is where the sign and the growth come *out*. Analyticity enters only through `normalized_families_force_germ_eq_at`, which supplies the local agreement (R) needs; (Z) and (R) themselves are C^2 statements. In (N) the regularity needed at the zeros is taken from analyticity, since `ContDiff.of_le` from ∞ to 2 has no one-line discharge on this pin while `AnalyticAt.contDiffAt` gives every grade for free.

3 Proof strategy

The anchor. Translate p to the origin, take S the closed unit ball, $a \equiv 1$, $m = 0$, $c = 1$ in `sector_lower_bound_multi`: on the scaled ball $t^{-1/2}S$ the bump is 1 once $t \geq 4/R^2$, the quadratic upper bound $K \leq C_0\|w\|^2$ comes from `quadratic_upper_bound_of_nonneg`, and `setIntegral_le_integral` compares with the full nonnegative integral. The constant is $\text{vol}(S)e^{-4C_0}$ and the exponent $-d/2$. An integer-power wrapper κt^{-d} for $t \geq 1$ matches the shape of `hanchor_low`.

The gap. $|\int \psi e^{-tL}| \leq \int |\psi| e^{-tL} \leq \|\psi\|_1 e^{-t\delta}$, pointwise on the support. For the product with a polynomially bounded B , Mathlib's `isLittle0.pow_exp_pos_mul_atTop` gives $t^n \leq e^{(\delta/2)t}$ eventually, so

$$|B \cdot I| \leq A\|\psi\|_1 e^{-(\delta/2)t},$$

and the seabed's `superPoly_of_eventually_abs_le_exp` finishes.

Tameness. At a zero p of L_1 with bump ψ_p ,

$$|C| I_1(\psi_p) \leq |I_2(\psi_p)| + |R_{\psi_p}| \leq M + 1$$

eventually, and the anchor gives $I_1(\psi_p) \geq \kappa t^{-d}$. At a zero q of L_2 : $C I_1(\psi_q) = I_2(\psi_q) - R_{\psi_q} \geq \frac{\kappa}{2} t^{-d}$ eventually, and $I_1(\psi_q) \leq M$, so $C \geq \frac{\kappa}{2M} t^{-d} > 0$.

Equal zero loci. If $L_1(p) = 0 < L_2(p)$, a bump in the gap of L_2 makes I_2 exponentially small, so $C I_1(\psi_p)$ is beyond all orders and the anchor makes C itself beyond all orders (`superPoly_of_mul_anchor` applied to $C + 1$); then every $I_2(\phi) = R_\phi + C I_1(\phi)$ is beyond all orders, contradicting the anchor at any zero of L_2 . If $L_2(q) = 0 < L_1(q)$, the upper tameness bound makes $C I_1(\psi_q)$ beyond all orders by (E), so $I_2(\psi_q)$ is, contradicting the anchor at q .

Rigidity and transfer. A bump ψ_0 inside the agreement neighbourhood has $I_1(\psi_0) = I_2(\psi_0)$ exactly (`anchor_moment_eq`), so $R_{\psi_0} = -(C - 1) I_1(\psi_0)$; the anchor and `superPoly_of_mul_anchor` give $C - 1 = o(t^{-\infty})$, and `superPoly_sub_of_scalar_gauge` against the bounded $I_1(\phi)$ removes the gauge from every tested moment.

The headline. Normalized agreement gives projective agreement with $C = Z_2/Z_1$ because Z_2 is bounded for $t \geq 0$ (the block of the morning's $1/Z$ theorem, now a standalone lemma

`projectiveAgreement_of_normalized`); then (Z), the locus form of the germ theorem at $W_0 = \{L_1 = 0\}$, and (R) at any zero.

4 Roadblocks and resolutions

None of substance. The two LSP rounds fixed: an implicit centre p the elaborator could not infer through the quadratic-bound lemma (passed as `(p := p)`); a `simp` that had already reduced 1^2 in the sector bound's integrand, so the `calc` step had to name the reduced form; a beta-redex in front of a rewrite inside `SuperPoly.congr` (the standing `beta_reduce` gotcha); and `ContDiff.of_le` (by `simp`) for $2 \leq \infty$, which does not fire on this pin. The last was resolved by design rather than by finding the coercion lemma: the closure theorems take C^2 at the zeros as a hypothesis, and the analytic corollaries discharge it with `AnalyticAt.contDiffAt`, which is what the germ theorem already assumes.

The consult, before writing, confirmed every statement and sharpened three things. On the scalar:

GPT-6 Astra: There is no issue with negative or nonmeasurable C . All estimates involving $C(t)$ are pointwise in t ; you never integrate C . In particular $|C(t)| I_1(\psi, t) \leq |I_2(\psi, t)| + |R_\psi(t)|$ gives the upper polynomial bound without a sign assumption.

On what the transfer does and does not give:

GPT-6 Astra: Thus the immediate conclusion is for C_c^∞ , not automatically C_c^0 . However, with globally smooth losses, the continuous-test conclusion is indeed true. [...] This proves local total-variation agreement beyond all orders, hence agreement against every continuous compactly supported observable. I would formalize that as a separate theorem.

And on the normalized/projective bridge:

GPT-6 Astra: For a fixed compactly supported window, nonnegative losses make Z_2 bounded, so normalized SuperPoly agreement implies projective agreement. The converse requires control of $1/Z_2$. A positive window supported away from all zeros can have exponentially small Z_2 .

The file follows all three: (T) has no sign hypothesis, the conclusions are stated for the tested class only, and (N) uses the one direction that holds.

5 What was Mathlib, what was new

Mathlib, all lookup:

- asymptotics: `isLittle0_pow_exp_pos_mul_atTop`;
- real powers: `Real.rpow_le_rpow_of_exponent_le`;
- bumps: the `ContDiffBump` API (`tsupport_eq`, `eventuallyEq_one`, `nonneg'`, `contDiff`, `hasCompactSupport`) and `Metric.eventually_nhds_iff`;

- integrals: `abs_integral_le_integral_abs` and `integral_mono`;
- set integrals: `setIntegral_le_integral` and `integral_add_left_eq_self`;
- supports: `image_eq_zero_of_notMem.tsupport`.

Seabed:

- the sector arc: `sector_lower_bound_multi`, `quadratic_upper_bound_of_nonneg`, and `lower_bound_not_superpolynomial`;
- the anchoring arc: `laplace_moment_bounded` and `anchor_moment_eq`;
- the gauge algebra: `superPoly_of_mul_anchor` and `superPoly_sub_of_scalar_gauge`;
- `superPoly_of_eventually_abs_le_exp`, the `SuperPoly` closure lemmas, and the morning's germ theorems.

New: the anchor in three forms, the gap bounds, the `ProjectiveAgreement` predicate and the `gapBump` helper, tameness, equal zero loci, scalar rigidity, the transfer, the analytic corollary, the standalone normalized-to-projective lemma, and the headline. About 650 lines; the anchor is a quarter of it.

6 Lessons

- For `CLAUDE.md`: `ContDiff.of_le` from ∞ to a finite grade has no by `simp` discharge on this pin. Where an analyticity hypothesis is in scope, `AnalyticAt.contDiffAt` is the route; otherwise take the finite grade as a hypothesis and let callers discharge it.
- For the tide skill: a hypothesis that several seabed theorems share and none discharges (`hanchor_low`, three consumers, one month) is a tide in itself. The survey step should grep for repeated undischarged hypotheses, not only for open questions in the paper.
- For the paper: the headline is “at a common analytic zero, projective agreement fixes its own gauge”. State the scope: modulo $o(t^{-\infty})$, for the tested class, and only in the presence of a zero on each side ($L_2 = L_1 + c$ is projectively identical to L_1 with $C = e^{-ct}$).

7 Follow-ups

- *Local total variation.* $\int_K |e^{-tL_2} - e^{-tL_1}| = o(t^{-\infty})$ for compact K , via $|a-b|^2 \leq t D(a-b)$ with $D = L_2 - L_1$ tested at the smooth observable $\eta^2 D$. Upgrades the transfer to continuous and bounded measurable observables. Needs smooth losses, not analyticity.
- *The converse bridge.* Projective agreement implies normalized agreement when the window is positive near a zero, by the anchor applied to Z_2 .
- *Retire the anchor hypothesis.* Restate the one-point anchoring contradiction and the packaged gauge removal of `Anchoring` with the lower bound discharged internally.

- *Two-sided tameness as one statement*, $ct^{-d/2} \leq C(t) \leq At^{d/2}$ with the sharp half-integer exponents; the file has the integer-exponent forms.
- *Paper*. Promote (Z), (R), (N) to the note as the closing theorems of the normalized section; Proposition 7.8 (one-point anchoring) becomes a corollary of (R).

Acknowledgements

GPT-6 Astra vetted every statement, insisted on the scope of the continuous-test upgrade and of the normalized-to-projective bridge, and proposed the tameness organisation. The seabed is the germbij singular arc, the anchoring tides, and the morning's normalized tide.